Lab 1 – Seeing Faces in Things: Pareidolia

CV8508 – Perceptual Cognition for Computer Vision

Hamilton et al. [1] showed that face detectors trained on human faces miss most faces that people see in objects (face pareidolia), and that training on animal faces or on pareidolic faces helps. You will test this on a small scale: a coding agent is allowed, and encouraged, for all code, but you must check what it does.

Setup

```
mkdir lab1 && cd lab1
wget https://utkarshmall13.github.io/cv8508-labs/lab1/setup.sh
sh setup.sh                        # downloads and unzips the data, builds .venv
source .venv/bin/activate
```

`data/` then contains (each folder has a `README.txt` with source and licence):

`human_faces` 5000 train / 500 val human face crops (UTKFace), 112 px.

`animal_faces` cat, dog and wild-animal faces (AFHQ), `train/` and `val/` with 700 / 150 images per class.

`backgrounds` non-face crops for `train/` and `val/`: `objects/` (Imagenette) and `web/` (random web photos).

`fit_test` Faces in Things test set: `crops/face/` has 927 pareidolic face crops and `crops/nonface/` 927 crops from outside the annotated boxes of the same photos; `fit_test.csv` has the metadata. *Never train on it.*

`fit_train` 400 more pareidolic face crops (`face/`) for training.

Tasks

Write each step as a script and keep it. Use the same seed, image size and training budget for all models.

1. **Human faces.** Train a ResNet-18 to separate human faces from backgrounds. Decide what to train on, how to measure performance, and report it on a held-out human split and on `fit_test`.
2. **Human + animal faces.** Retrain with human and animal faces as the positive class. Report on a mixed held-out split and on `fit_test`.
3. **Pareidolic faces.** Train a third ResNet-18 on `fit_train` faces vs. backgrounds. Report on `fit_test`.
4. **You vs. the models.** Randomly sample 50 crops from `fit_test` (faces and non-faces pooled, fixed seed) and save them *without their labels*. Annotate them yourself (face / no face), then compute your accuracy. Next, score the same 50 images with the three models and report each model's accuracy at the threshold that gives the highest F1, next to yours.

Analysis

Answer in a few sentences each, citing your own numbers, and mark each claim as evidence, hypothesis or speculation.

1. Which metric did you choose, and why? Why is a threshold-free metric useful here, and when is accuracy at a threshold still what you need?
2. Compare the three models on `fit_test`. What do the differences say about what "face" means to each one?
3. How accurate were you on the 50 crops, and what cues did you use? Which crops did you and the models get wrong?
4. At the best-F1 threshold, how do the models compare with you? With only 50 images, how big a gap would be meaningful?
5. How could the test mislead (cropping of negatives, unannotated faces, tuning the threshold on the same 50 images)?
6. Propose one experiment that would show whether animal faces specifically help.

References

[1] Mark Hamilton, Simon Stent, Vasha DuTell, Anne Harrington, Jennifer Corbett, Ruth Rosenholtz, and William T. Freeman. Seeing faces in things: a model and dataset for pareidolia. In *European Conference on Computer Vision (ECCV)*, 2024.